

Ordering Rational Numbers on the Number Line

Jonathan R. Bacolod

Rational numbers include integers, fractions, decimals, and terminating or repeating decimals.

Number Line

On a number line, numbers farther to the **left** are smaller, while numbers farther to the **right** are larger.

smaller ← • → larger

Positive and Negative Numbers

Negative numbers are always less than positive numbers.

$$-5 < 2$$

A number line helps visualize the relative positions of rational numbers.

Comparing Negative Numbers

Among negative numbers, the number with the greater **absolute value** is actually smaller.

$$-8 < -3$$

On the number line, -8 is farther to the left than -3 .

Comparing Fractions

To compare **fractions**, rewrite them with a common denominator or convert them to decimals.

$$\frac{1}{2} = 0.5 \quad \frac{3}{4} = 0.75$$

Therefore,

$$\frac{1}{2} < \frac{3}{4}$$

Ordering Rational Numbers

To order rational numbers:

1. Compare their values.
2. Arrange them from least to greatest or greatest to least.

For example,

$$-2 < -\frac{1}{2} < 0 < \frac{3}{4}$$

Examples

Example 1

Arrange -3 , 2 , -1 , 5 from least to greatest.

$$-3 < -1 < 2 < 5$$

Example 2

Arrange $-\frac{1}{2}$, $\frac{3}{4}$, $-\frac{5}{4}$, 0 from least to greatest.

$$-\frac{5}{4} < -\frac{1}{2} < 0 < \frac{3}{4}$$

Example 3

Arrange 1.2, -0.8 , 0, -1.5 from least to greatest.

$$-1.5 < -0.8 < 0 < 1.2$$

Practice Exercises

Practice 1

Arrange the following from least to greatest.

$$-3, 2, -1, 5$$

Practice 2

Arrange the following from least to greatest.

$$-\frac{1}{2}, \frac{3}{4}, -\frac{5}{4}, 0$$

Practice 3

Arrange the following from least to greatest.

$1.2, -0.8, 0, -1.5$

Practice 4

Arrange the following from least to greatest.

$$\frac{2}{3}, -\frac{1}{3}, \frac{5}{6}, -\frac{3}{2}$$

Practice 5

Arrange the following from least to greatest.

$$-4, -2.5, -3, -1$$

Practice 6

Arrange the following from least to greatest.

$$\frac{7}{5}, 1, -\frac{2}{5}, \frac{1}{5}$$

Practice 7

Arrange the following from least to greatest.

$$-0.25, \frac{1}{2}, -\frac{3}{4}, 0.75$$

Practice 8

Arrange the following from least to greatest.

$$\frac{5}{8}, \frac{1}{4}, -\frac{3}{8}, -\frac{1}{8}$$

Practice 9

Arrange the following from least to greatest.

$$-2.1, -2, -1.9, -2.3$$

Practice 10

Arrange the following from least to greatest.

$$\frac{9}{10}, -\frac{7}{10}, \frac{1}{10}, -\frac{2}{10}$$

Activities

Activity 1

Plot the given rational numbers on a number line, then write them from least to greatest.

$$-4, 3, -2, 6$$

Activity 2

Plot the given rational numbers on a number line, then write them from least to greatest.

$$-\frac{3}{4}, \frac{1}{2}, -\frac{5}{2}, 0$$

Activity 3

Plot the given rational numbers on a number line, then write them from least to greatest.

$1.8, -0.4, 0, -1.2$

Activity 4

Plot the given rational numbers on a number line, then write them from least to greatest.

$$\frac{3}{4}, -\frac{1}{4}, \frac{7}{8}, -\frac{5}{4}$$

Activity 5

Plot the given rational numbers on a number line, then write them from least to greatest.

$$-5, -2, -3.5, -1.5$$

Activity 6

Plot the given rational numbers on a number line, then write them from least to greatest.

$$\frac{8}{5}, 1, -\frac{1}{5}, \frac{2}{5}$$

Activity 7

Plot the given rational numbers on a number line, then write them from least to greatest.

$$-0.5, \frac{1}{4}, -\frac{7}{8}, 0.625$$

Activity 8

Plot the given rational numbers on a number line, then write them from least to greatest.

$$\frac{3}{8}, \frac{7}{8}, -\frac{1}{4}, -\frac{5}{8}$$

Activity 9

Plot the given rational numbers on a number line, then write them from least to greatest.

$$-3.2, -3, -2.8, -3.5$$

Activity 10

Plot the given rational numbers on a number line, then write them from least to greatest.

$$\frac{8}{10}, -\frac{6}{10}, \frac{2}{10}, -\frac{1}{10}$$

Thank you!